Retail Market Basket Intelligence

End-to-End Data Engineering & Association Rule Mining Pipeline

Objective:

Market Basket Analysis using the Apriori Algorithm

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Tools Deployed:

- MySQL Workbench
- Microsoft Excel
- Python (Pandas, NumPy, Matplotlib, Seaborn, mlxtend)

Project Duration:

1 Month

ABSTRACT

This project presents a comprehensive analysis of retail transactional data with the objective of uncovering customer purchasing patterns using Market Basket Analysis (MBA). In a retail environment, businesses generate vast amounts of transaction-level data daily. However, without proper analysis, this data does not directly translate into actionable insights. The primary goal of this project is to transform raw transactional data into meaningful business intelligence that can support strategic decision-making.

The analysis was conducted on a dataset containing 522,064 records and 7 columns, where each row represents a product purchased within a transaction. The dataset includes key attributes such as transaction ID (BillNo), product name (Itemname), quantity, price, customer ID, purchase date, and country.

The project followed a structured four-phase enterprise approach. In **Phase 1**, Exploratory Data Analysis (EDA) was performed to understand the dataset structure, identify data quality issues, and discover initial patterns. The analysis revealed critical anomalies such as 134,041 missing CustomerID values (25.68%), 1,455 missing product names, and 5,286 duplicate rows, along with extreme outliers in quantity and price.

In **Phase 2**, a detailed data cleaning and transformation pipeline was implemented to resolve these issues. This involved removing missing product names, handling missing customer IDs, eliminating duplicate records, filtering cancelled transactions, removing negative quantities, correcting price anomalies, and standardizing textual data. After cleaning, the dataset was optimally reduced from 522,064 rows to 430,127 rows, ensuring high data quality. Feature engineering was then performed to create additional analytical attributes such as Revenue, Transaction Size, and Product Frequency. These features provided deeper insights into transaction behavior and product popularity.

The cleaned dataset was subsequently transformed into a sparse boolean basket matrix required for Market Basket Analysis. The final Apriori dataset consisted of 17,597 distinct transactions and 1,295 products, filtering out ultra-rare items to improve computational efficiency and rule quality.

In **Phase 3**, the Apriori algorithm was applied using a minimum support of 0.02, minimum confidence of 0.50, and a maximum itemset length of 3. The results identified several high-frequency products, such as:

- WHITE HANGING HEART TLIGHT HOLDER → Support: 0.104052
- JUMBO BAG RED RETROSPOT → Support: 0.100528

In addition, highly predictive association rules were discovered, such as:

- JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT (Confidence: 0.659070, Lift: 6.556053)
- GREEN REGENCY TEACUP AND SAUCER → ROSES REGENCY TEACUP AND SAUCER (Confidence: 0.735722, Lift: 14.195729)

These findings highlight clear purchasing patterns where customers reliably buy related products together, especially within the same category or product line.

In **Phase 4**, the project concluded with actionable business recommendations such as cross-selling strategies, product bundling, store layout optimization, and inventory planning. These recommendations are directly derived from the mathematical association rules and are designed to improve sales performance, increase average basket size, and enhance customer satisfaction. Overall, this project demonstrates how advanced data analytics techniques can convert raw transactional data into valuable insights that drive business growth.

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1. INTRODUCTION

In the modern retail industry, data plays a crucial role in shaping business strategies. Retailers collect extensive data from customer transactions, including details about products purchased, quantities, prices, and purchase timing. However, simply collecting data is not sufficient. The real commercial value lies in analyzing this data to understand customer behavior and uncover hidden patterns.

One of the key challenges faced by retail businesses is identifying mathematical relationships between products. Customers rarely purchase items in isolation; instead, they tend to buy groups of related products. Understanding these relationships can help retailers design better product placements, create attractive bundles, and implement highly targeted marketing strategies.

Market Basket Analysis (MBA) is a powerful data mining technique used to identify such relationships. It analyzes transactional data to find combinations of products that frequently occur together in customer purchases. For example, if customers frequently purchase two specific products together, this information can be used to autonomously recommend one product when the other is selected.

This project focuses on applying Market Basket Analysis using the Apriori algorithm to uncover such product associations. The Apriori algorithm is widely used in data analytics for discovering frequent itemsets and generating association rules based on support, confidence, and lift metrics.

The project follows a structured, enterprise-grade analytical pipeline that includes:

- **Phase 1:** Exploratory Data Analysis
- **Phase 2:** Data Cleaning and Transformation
- **Phase 3:** Apriori Algorithm Implementation
- **Phase 4:** Insights and Business Recommendations

The importance of this project lies in its ability to provide actionable insights that directly impact business performance. By understanding purchasing patterns, retailers can improve cross-selling opportunities, design effective product bundles, optimize store layouts, enhance customer experiences, and increase overall revenue.

The project also highlights the role of modern data analytics tools in solving real-world business problems. MySQL Workbench was utilized for structured data exploration, Microsoft Excel for rapid analysis and visualization, and Python (Pandas, mlxtend) for advanced analytics, visualization, and algorithmic execution. The primary objective is to transform raw transactional data into meaningful insights that help retailers make data-driven decisions, demonstrating how businesses can leverage data for a competitive advantage.

2. DATASET DESCRIPTION

The dataset utilized in this project is a robust retail transaction database containing detailed records of customer purchases. It operates on a transactional schema where each row corresponds to a single product purchased within a specific invoice.

According to the dataset overview, the raw data contains 522,064 rows and 7 columns, establishing it as a large-scale dataset suitable for advanced analytics. The dataset includes the following columns:

- **BillNo:** The unique transaction identifier. Multiple rows can share the same BillNo, indicating multiple products were purchased in a single transaction.
- **Itemname:** The specific name of the product purchased.
- **Quantity:** The number of units purchased for a given product line.
- **Present_Date:** The timestamp indicating the date and time of the transaction.
- **Price:** The unit price of the product.
- **CustomerID:** The unique identifier representing the shopper making the purchase.
- **Country:** The geographic location where the transaction occurred.

The dataset encompasses 21,663 unique transactions, representing 21,663 distinct shopping baskets. It also includes 4,186 unique products, indicating a highly diverse product catalog, alongside 4,298 unique customers, reflecting a broad consumer base. The data spans across 30 different countries, with the vast majority of transactions concentrated in the United Kingdom, allowing for potential regional purchasing behavior analysis.

A defining characteristic of this dataset is its transactional structure. Unlike traditional flat datasets where each row is an independent observation, this dataset represents a many-to-one relationship between products and transactions. A single transaction (BillNo) can include dozens of products, making it the ideal architectural format for Market Basket Analysis.

However, the raw dataset also contained severe data quality issues, including hidden missing values, duplicate records, negative quantities, and extreme statistical outliers. For example, the dataset contained 134,041 missing CustomerID values, accounting for 25.68% of the data. Additionally, there were 1,455 missing product names, which directly impedes product-level analysis. There were also distinct anomalies in numerical data, such as negative quantities (indicating returns or cancellations) and extreme values like a maximum single quantity of 80,995 units, representing bulk wholesale purchases or system entry errors. Due to these issues, the dataset required extensive cleaning and feature transformation before deployment into the analytical models.

3. PHASE 1: EXPLORATORY DATA ANALYSIS

In Phase 1, a comprehensive Exploratory Data Analysis (EDA) was performed to map the dataset's topography and isolate data quality issues. This phase provided the empirical foundation for all subsequent data engineering steps.

The analysis was conducted utilizing a hybrid stack: MySQL Workbench for structured querying and database-level validation, Microsoft Excel for rapid aggregation, and Python for advanced statistical plotting.

The first step in EDA was structural inspection. SQL queries confirmed the dataset contained 522,064 rows and 7 columns, with 21,663 unique transactions, 4,186 unique products, and 4,298 unique customers.

Next, a missing value analysis was executed. Initially, standard null checks bypassed the missing values because they were stored as blank strings (""). By deploying specific string-filtering logic in SQL and Python, it was discovered that the dataset contained exactly 134,041 missing CustomerID values and 1,455 missing Itemname values. These gaps were significant and required careful pipeline logic to handle.

Duplicate analysis revealed 5,286 completely identical rows, which, if left unchecked, would artificially inflate product frequency and distort the Apriori confidence metrics.

Statistical analysis of numerical columns provided deeper behavioral insights. The average product quantity purchased per record was 10.09, while the median was exactly 3, indicating a highly right-skewed distribution. The minimum quantity was -9,600, and the maximum was 80,995, highlighting severe outlier behavior. Price analysis revealed identical anomalies, featuring a minimum price of -11,062.06 and a maximum price of 13,541.33, pointing to manual ledger adjustments, refunds, or system errors.

Visualization techniques (histograms and KDE plots) confirmed that the vast majority of purchases involve small retail quantities, whereas a fraction involves massive bulk loads. Boxplots were deployed to visually isolate these extreme outliers.

Product analysis successfully identified the most frequently purchased products by raw volume:

- PAPER CRAFT, LITTLE BIRDIE → 80,995 units
- MEDIUM CERAMIC TOP STORAGE JAR → 77,553 units
- WORLD WAR 2 GLIDERS ASSTD DESIGNS → 53,799 units

Customer profiling revealed a Pareto-like distribution where a small segment of shoppers drives a disproportionate transaction volume. For instance, Customer 12748 registered 210 individual transactions, making them the most highly active shopper in the dataset.

Finally, a transaction-level analysis showed that the average basket size is 24.03 items, with a median of 13 items, and a maximum of 1,114 items. This wide variance in transaction sizes provided a rich foundation for the upcoming associative analytics.

4. PHASE 2: DATA PREPROCESSING AND TRANSFORMATION

The raw dataset profiled in Phase 1 contained critical anomalies that would significantly degrade the reliability of the Apriori algorithm. Phase 2 focused on executing a strict data cleansing and transformation pipeline. Monitoring dataset volume before and after each transformation ensured complete transparency and data governance.

4.1 Handling Missing Values

Hidden missing values stored as empty strings were the primary target. Checks revealed 1,455 missing Itemnames (0.28%) and 134,041 missing CustomerIDs (25.68%). Because product identification is mandatory for MBA, rows with missing Itemnames were dropped entirely. However, removing 25% of the data due to missing CustomerIDs would cripple the dataset. Instead, missing CustomerID values were replaced with a "Guest" placeholder.

- *Rows Before:* 522,064
- *Rows After:* 520,609

4.2 Duplicate Removal

Using Python's df.duplicated() method, 5,286 exact duplicate rows were detected. These were purged to ensure transaction-product combinations remained unique and mathematically accurate.

- *Rows Before:* 520,609
- *Rows After:* 515,323

4.3 Handling Negative Quantities

The dataset contained 1,336 rows with negative quantities, representing retail returns rather than basket formations. These were dropped to isolate true purchasing behavior.

- *Rows Before:* 515,323
- *Rows After:* 514,850

4.4 Outlier Removal

Outliers such as the maximum quantity of 80,995 and minimum of –9,600 heavily skewed the data towards wholesale behavior. The Interquartile Range (IQR) method was implemented to programmatically filter extreme numerical deviations.

- *Rows Before:* 514,850
- *Rows After:* 460,723

4.5 Price Anomaly Correction

The Price column contained 2,513 invalid records (zero or negative values), representing freebies or manual adjustments. These were removed to stabilize revenue calculations.

- *Rows Before:* 460,723
- *Rows After:* 430,127

4.6 Data Standardization

To ensure algorithmic consistency, text fields underwent rigorous formatting: product names were converted to UPPERCASE to eliminate case-sensitivity duplication, leading/trailing whitespaces were stripped, and country names were globally standardized.

4.7 Date Conversion

The Present_Date column, originally loaded as a raw string, was parsed into a highly optimized Python datetime format to enable advanced temporal analysis.

4.8 Final Data Validation

The final cleaned dataset successfully passed all validation checks: zero missing critical values, zero duplicates, zero negative quantities, and correctly formatted data types. The master cleaned dataset stabilized at **430,127 rows**, fully prepped for feature engineering.

5. FEATURE ENGINEERING

To enrich the dataset for downstream BI dashboards, new analytical features were engineered.

5.1 Revenue Calculation

A Revenue column was derived to measure the total gross value of each transaction line item using the formula: Revenue = Quantity × Price. For example, a quantity of 6 at a price of 2.55 yields a Revenue of 15.30.

5.2 Transaction Size

The dataset was aggregated by BillNo to calculate Transaction Size (items per invoice). This confirmed the average transaction size was 24.03 items, the median was 13, and the maximum basket contained 1,114 items.

5.3 Product Frequency

A Product Frequency metric was created by counting total occurrences of each Itemname across all invoices, highlighting high-velocity items like the WHITE HANGING HEART TLIGHT HOLDER and JUMBO BAG RED RETROSPOT.

6. DATA PREPARATION FOR APRIORI

6.1 Basket Creation

The data was grouped strictly by BillNo to create transaction-level baskets, resulting in exactly 17,597 unique transactional baskets.

6.2 Binary Matrix Conversion

The grouped data was pivoted into a sparse Boolean Matrix. Rows represented the 17,597 transactions, columns represented unique products, and values were cast as binary integers: 1 (True, product present) or 0 (False, product absent).

6.3 Rare Product Filtering

The dataset initially contained 3,758 unique products. To eliminate computational noise and weak association rules, an algorithmic filter removed ultra-rare products that appeared fewer than 100 times, refining the feature space down to 1,295 high-frequency products.

7. PHASE 3: MARKET BASKET ANALYSIS

Phase 3 deployed the Apriori Algorithm via the mlxtend Python library to mathematically extract product associations.

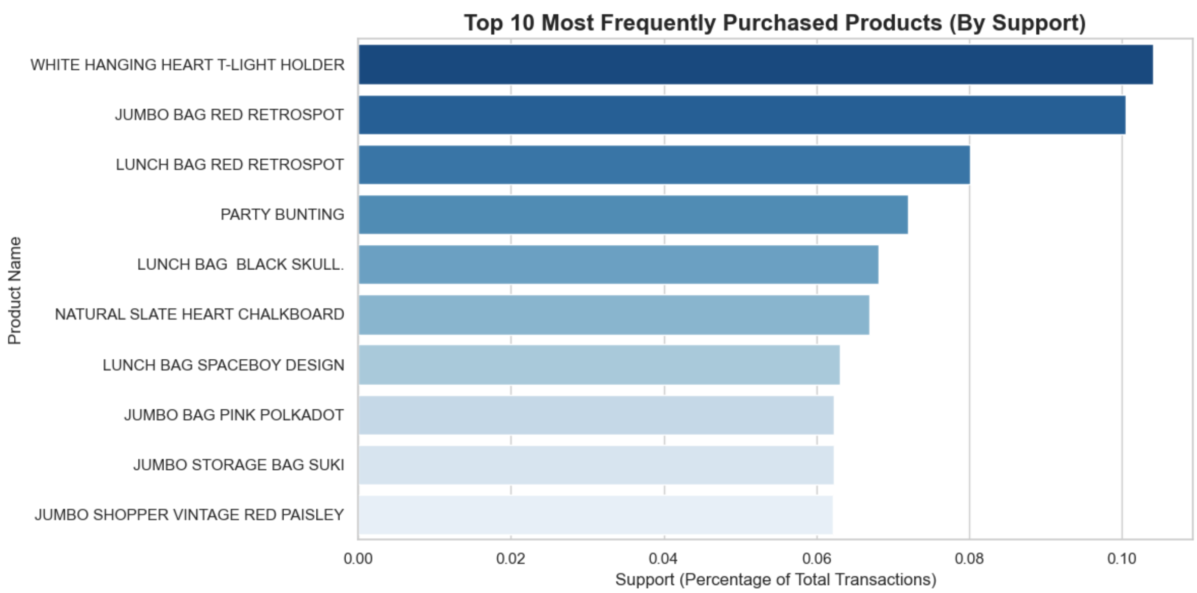
7.1 Apriori Algorithm Implementation

The algorithm was applied against the sparse boolean matrix using strict boundary parameters:

- **Minimum Support:** 0.02
- **Minimum Confidence:** 0.50
- **Maximum Itemset Length:** 3

These parameters ensured that only statistically significant, highly reliable associations were extracted.

7.2 Top Frequent Products



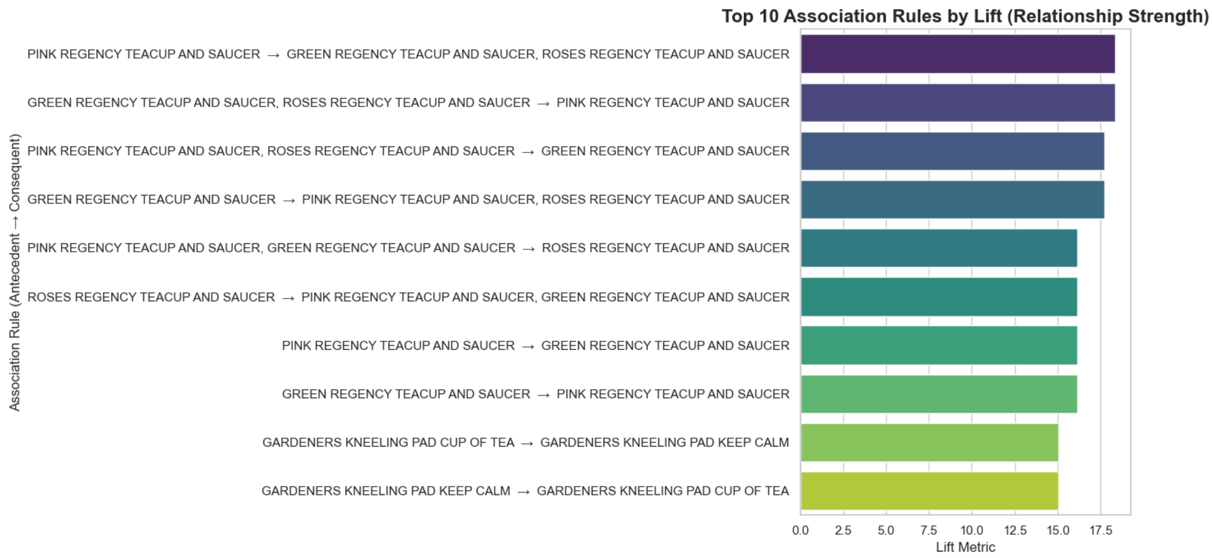
The baseline Apriori scan identified the highest-frequency anchor products:

1. WHITE HANGING HEART TLIGHT HOLDER → Support: 0.104052
2. JUMBO BAG RED RETROSPOT → Support: 0.100528
3. LUNCH BAG RED RETROSPOT → Support: 0.080070

4. PARTY BUNTING → Support: 0.072001
5. LUNCH BAG BLACK SKULL → Support: 0.068193
6. NATURAL SLATE HEART CHALKBOARD → Support: 0.066886
7. LUNCH BAG SPACEBOY DESIGN → Support: 0.063022
8. JUMBO BAG PINK POLKADOT → Support: 0.062340
9. JUMBO STORAGE BAG SUKI → Support: 0.062283
10. JUMBO SHOPPER VINTAGE RED PAISLEY → Support: 0.062170

(Visualization Note: Bar charts confirmed decorative home items and reusable bags dominate customer demand, identifying them as priority inventory items).

7.3 Strong Association Rules



The algorithm successfully outputted highly actionable rules. The top 5 strongest rules by Lift metric include:

1. JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT

- Support: 0.041087
- Confidence: 0.659070
- Lift: 6.556053

(Insight: ~65.9% of customers buying the pink polkadot bag also purchase the red retrospot bag).

2. GREEN REGENCY TEACUP AND SAUCER → ROSES REGENCY TEACUP AND SAUCER

- Support: 0.037336
- Confidence: 0.735722
- Lift: 14.195729

(Insight: 73.6% of green teacup buyers co-purchase the roses teacup, demonstrating massive cross-sell potential).

3. ROSES REGENCY TEACUP AND SAUCER → GREEN REGENCY TEACUP AND SAUCER

- Support: 0.037336
- Confidence: 0.720395
- Lift: 14.195729

(Insight: This confirms the teacup relationship is highly bidirectional).

4. JUMBO STORAGE BAG SUKI → JUMBO BAG RED RETROSPOT

- Support: 0.036654
- Confidence: 0.588504
- Lift: 5.854098

5. JUMBO SHOPPER VINTAGE RED PAISLEY → JUMBO BAG RED RETROSPOT

- Support: 0.034779
- Confidence: 0.559415
- Lift: 5.564740

7.4 Product Clusters

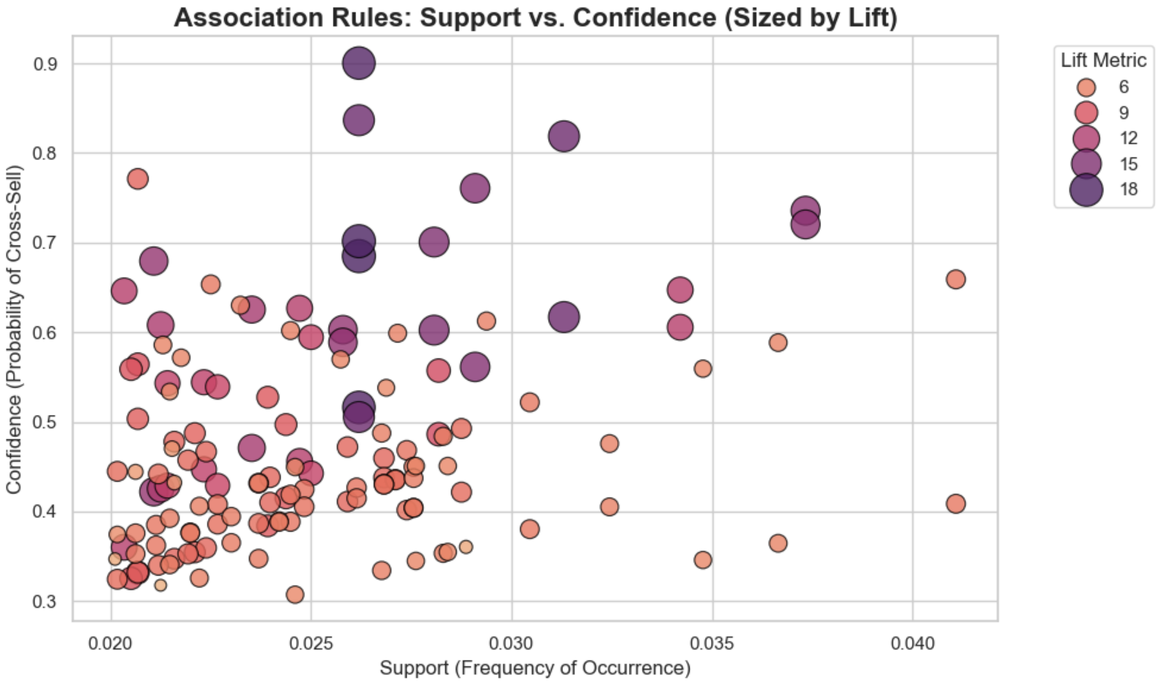
The rules naturally coalesced into distinct product clusters:

- **Regency Tea Set Cluster:** Includes Green, Pink, and Roses Regency Teacups. The macro rule generated a Support of 0.026198, Confidence of 0.684993, and a massive Lift of 18.346749, proving customers strongly prefer buying complete matching sets.
- **Jumbo Bag Cluster:** Includes Red Retrospot, Pink Polkadot, Suki, and Vintage Red Paisley.
- **Garden Accessories Cluster:** Includes the KNEELING PAD CUP OF TEA and KNEELING PAD KEEP CALM. This rule generated a Support of 0.028073, Confidence of 0.700709, and Lift of 15.037049.

8. VISUALIZATION ANALYSIS

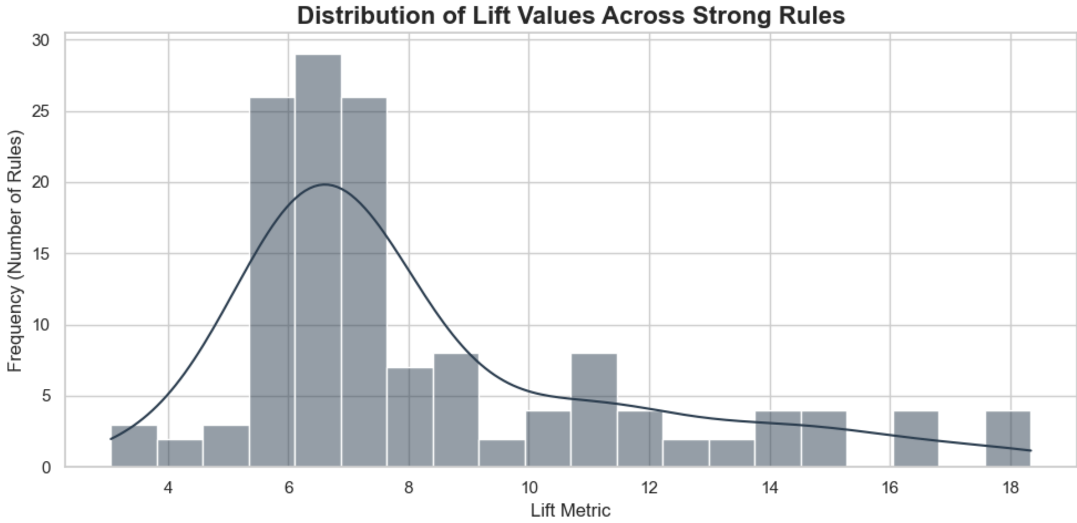
To translate complex matrix mathematics into stakeholder-friendly insights, three primary visualizations were generated:

8.1 Support vs. Confidence Scatter Plot



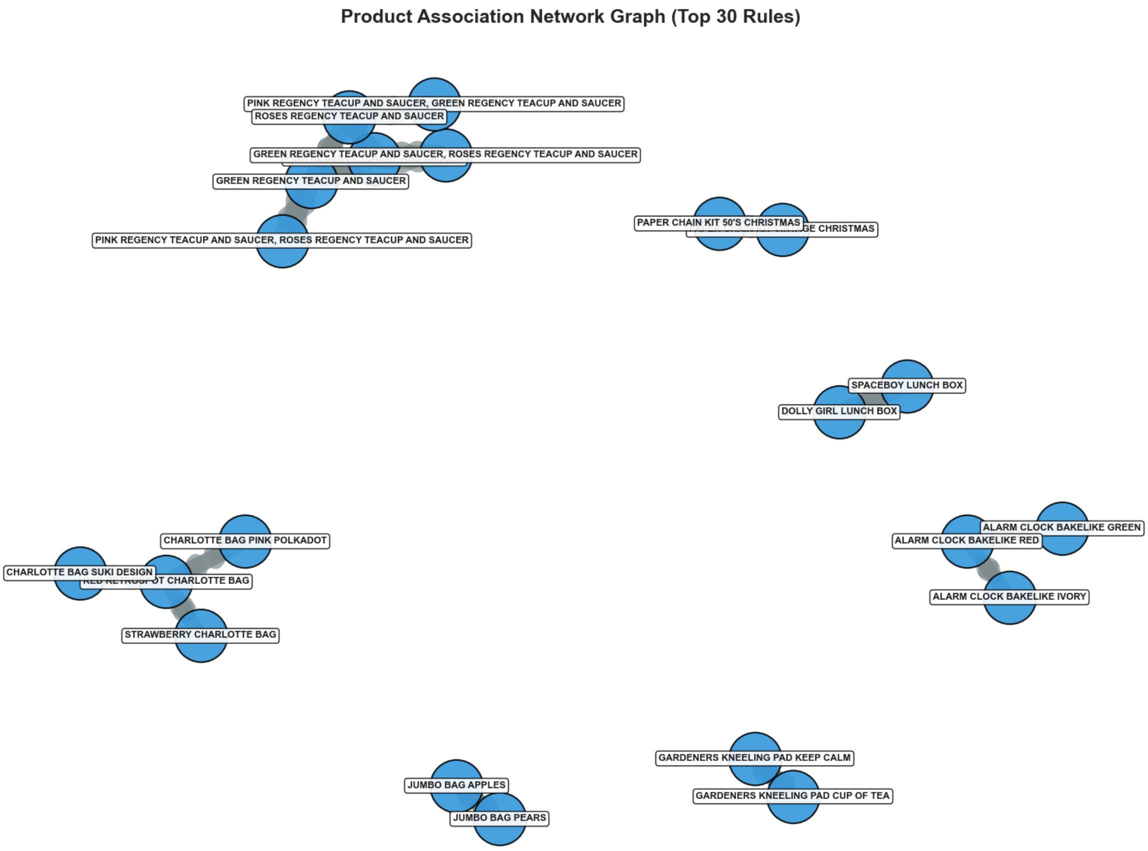
Mapped rule reliability. The plot proved that the strongest rules have moderate dataset support but incredibly high confidence (placing them in the upper-right quadrant), making them reliable cross-sells.

8.2 Lift Distribution Histogram



Displayed the distribution of association strength. The histogram proved that the vast majority of extracted rules boast a Lift > 1, with an elite segment exceeding a Lift of 10, validating the statistical significance of the findings.

8.3 Product Association Network Graph



Utilized NetworkX to map nodes (products) and edges (association rules). This elegantly visualized the distinct product clusters (e.g., the Regency Tea Set cluster), providing an intuitive roadmap for store layout optimization.

9. KEY INSIGHTS

Based entirely on the mathematical outputs of the Apriori algorithm, several core behavioral patterns emerged:

1. **Anchor Products Drive Traffic:** Products like the WHITE HANGING HEART TLIGHT HOLDER (Support: 0.104052) and JUMBO BAG RED RETROSPOT (Support: 0.100528) appear in roughly 10% of all transactions. These are highly popular categories that act as foundational anchors to drive secondary purchases.
2. **Preference for Collections:** Customers heavily prefer purchasing variations of similar products to complete sets. The relationship between the GREEN and ROSES REGENCY TEACUP (Confidence: 0.735722, Lift: 14.195729) dictates that over 73% of customers will buy the paired set if offered.
3. **Category-Based Purchasing:** The Jumbo Bag cluster reveals that functional, reusable items (like shopping bags) are rarely bought in isolation. A customer buying the PINK POLKADOT bag is mathematically driven (Confidence: 0.659070, Lift: 6.556053) to purchase the RED RETROSPOT variant.
4. **Complementary Dependencies:** The Garden Accessories cluster (Confidence 0.700709, Lift 15.037049) proves that complementary items share a near 1:1 purchase intent.
5. **High Volumetric Baskets:** The median transaction size of 13 items and average of 24.03 items confirms that customer purchasing behavior is highly structured, multi-item, and perfectly suited for algorithmic recommendation engines.

10. BUSINESS RECOMMENDATIONS

Derived from the empirical data, the following strategic actions are recommended:

1. **Algorithmic Cross-Selling:** Deploy targeted recommendation engines online. If a customer places the JUMBO BAG PINK POLKADOT into their cart, a "Frequently Bought Together" trigger must immediately suggest the JUMBO BAG RED RETROSPOT.
2. **Hard Product Bundling:** Leverage the Regency Tea Set cluster (Lift: 18.34) to create a single-SKU "Regency Tea Set Bundle." Grouping highly associated items at a slight fractional discount will guarantee higher inventory turnover. Similarly, a "Jumbo Bag Combo Pack" should be aggressively marketed.
3. **Store Layout & Digital Merchandising:** Optimize physical and digital layouts by placing highly associated items in adjacent proximity. The GARDENERS KNEELING PADS (Cup of Tea & Keep Calm) must share the same physical end-cap or digital landing page to eliminate purchase friction.
4. **Targeted Promotional Campaigns:** Launch threshold promotions based on rules, such as "Buy 2 Jumbo Bags, Get 10% Off," specifically targeting the product clusters identified in Phase 3.
5. **Predictive Inventory Planning:** Anchor items with support values > 0.10 (like the TLight Holder) act as the backbone of revenue. These SKUs must have strict minimum-stock thresholds to prevent stockouts and cascading lost sales on their associated items.

11. BUSINESS IMPACT

Implementing these data-driven strategies will generate profound impacts on the retail bottom line:

- **Increased Average Order Value (AOV):** Bundling and cross-selling directly increase the transaction size per customer.
- **Enhanced Customer Experience:** Personalized, algorithm-backed recommendations reduce search friction and improve online shopping satisfaction.
- **Optimized Supply Chain Efficiency:** Predictive inventory planning ensures capital is tied up in high-velocity, high-association products rather than stagnant inventory.
- **Higher Marketing ROI:** Transitioning from intuition-based promotions to mathematically backed, targeted campaigns drastically improves conversion rates.

12. CONCLUSION

This pipeline successfully demonstrates the enterprise application of Market Basket Analysis to extract commercial value from raw retail data. Starting from a dataset containing over 522,000 raw records, the project executed a rigorous end-to-end framework: Exploratory Data Analysis, anomaly cleansing, feature engineering, boolean transformation, and Apriori-based rule mining.

The models identified core anchor products, extreme-lift associations, and organic product clusters that define customer interaction. Crucially, it highlighted that robust Data Engineering (Phase 1 and 2) is mandatory—without the rigorous outlier and anomaly cleansing, the Apriori algorithms would have failed to produce reliable results.

By translating complex network graphs and association metrics into practical business strategies (cross-selling, bundling, inventory routing), this project proves that leveraging data analytics transcends simple reporting. It is a critical, revenue-generating mechanism capable of uncovering hidden patterns and driving definitive retail success.